

Signals & System

Lecture 19

Discrete-Time Transfer Functions

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Discrete-Time Transfer Functions

Transfer Function of a Discrete-Time Systems (2 lectures): Impulse sampler, Laplace transform of impulse sequence, z transform. **Properties of the z transform. Examples. Difference equations and differential equations. Digital filters.**

Specific objectives for today:

- Properties of the z-transform
- z-transform transform equations
- Solving difference equations

Lecture 19: Resources

Core material

SaS, O&W, C10

Related Material

MIT lecture 22 & 23

The **discrete time transfer function** (z-transform), closely mirrors the Laplace transform (continuous time transfer function) and is the **z-transform** of the **impulse response of the difference equation**.

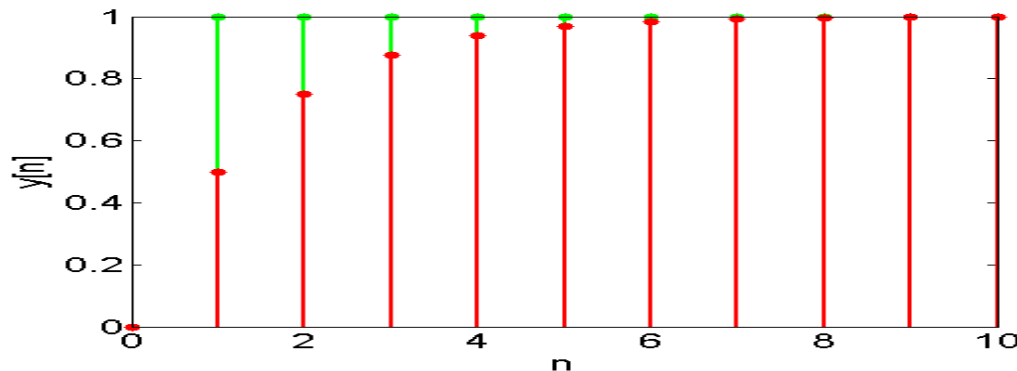
Introduction to Discrete Time Transfer Fns

A **discrete-time LTI system** can be represented as a (first order) **difference equation** of the form:

$$a_1 y[n] + a_2 y[n-1] = b_1 x[n] + b_2 x[n-1]$$

$$y[n] = (-a_2 y[n-1] + b_1 x[n] + b_2 x[n-1]) / a_1$$

This is analogous to a sampled CT differential equation



This is hard to solve analytically, and we'd like to be able to perform some form of analogous manipulation like continuous time **transfer functions**, i.e.

$$Y(s) = H(s)X(s)$$

Linearity of the z-Transform

If $x_1[n] \xleftrightarrow{z} X_1(z)$ ROC = R_1

and $x_2[n] \xleftrightarrow{z} X_2(z)$ ROC = R_2

Then $ax_1[n] + bx_2[n] \xleftrightarrow{z} aX_1(z) + bX_2(z)$ ROC = $R_1 \cap R_2$

This follows directly from the definition of the z-transform (as the summation operator is linear, see Example 3). It is easily extended to a linear combination of an arbitrary number of signals

Time Shifting & z-Transforms

If $x[n] \xleftrightarrow{Z} X(z)$ ROC=R

Then $x[n - n_0] \xleftrightarrow{Z} z^{-n_0} X(z)$ ROC=R

Proof

$$\begin{aligned} Z\{x[n-1]\} &= \sum_{n=-\infty}^{\infty} x[n-1]z^{-n} \\ &= z^{-1} \sum_{n=-\infty}^{\infty} x[n-1]z^{-(n-1)} \\ &= z^{-1} \sum_{m=-\infty}^{\infty} x[m]z^{-m} = z^{-1} Z\{x[n]\} \end{aligned}$$

This is very important for producing the **z-transform transfer function of a difference equation** which uses the property:

$$x[n-1] \xleftrightarrow{Z} z^{-1} X(z)$$

Example: Linear & Time Shift

Consider the input signal

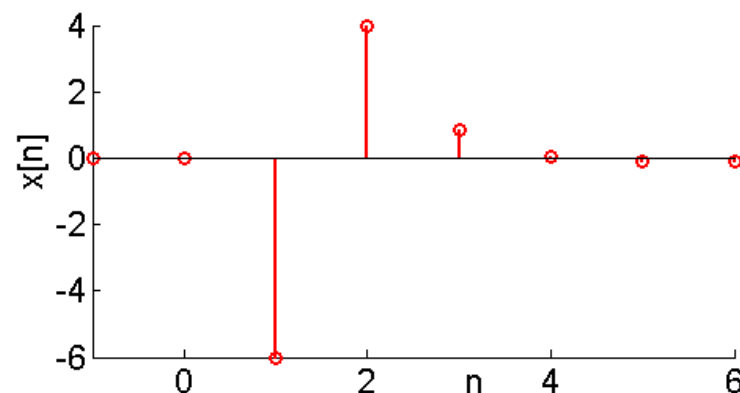
$$x[n] = 7(1/3)^{n-2}u[n-2] - 6(1/2)^{n-1}u[n-1]$$

We know that

$$a^n u[n] \xleftrightarrow{z} \frac{z}{z-a}$$

So

$$\begin{aligned} X(z) &= 7z^{-2} \frac{z}{z-1/3} - 6z^{-1} \frac{z}{z-1/2} \\ &= 7 \frac{1}{z^2 - 1/3z} - 6 \frac{1}{z - 1/2} \end{aligned}$$



Discrete Time Transfer Function

Consider a first order, LTI differential equation such as:

$$a_1 y[n] + a_2 y[n-1] = b_1 x[n] + b_2 x[n-1]$$

Then the **discrete time transfer function** is the **z-transform of the impulse response, $H(z)$**

As $Z\{\delta[n]\} = 1$, taking the z-transform of both sides of the equation (linearity we get), for the impulse response

$$Z\{a_1 h[n] + a_2 h[n-1]\} = Z\{b_1 \delta[n] + b_2 \delta[n-1]\}$$

$$(a_1 + a_2 z^{-1})Z\{h[n]\} = (b_1 + b_2 z^{-1})Z\{\delta[n]\}$$

$$\begin{aligned} H(z) &= \frac{(b_1 + b_2 z^{-1})}{(a_1 + a_2 z^{-1})} \\ &= \frac{(zb_1 + b_2 z)}{(za_1 + a_2)} \end{aligned}$$

Discrete Time Transfer Function

The discrete-time transfer function of an LTI system is a **rational polynomial in z** . (This is equivalent to the transfer function of a continuous time differential system which is a rational polynomial in s)

$$H(z) = \frac{(zb_1 + b_2)}{(za_1 + a_2)}$$

As usual the z -transform transfer function can be computed by either:

1. If the difference equation is known, take the z -transform of each side when the input signal is an impulse $\delta[n]$
2. If the discrete-time impulse response signal is known, calculate the z -transform of the signal $h[n]$.

In either case, the same result will be obtained.

Convolution using z-Transforms

The z-transform also has the multiplication property, i.e.

$$x[n] \xleftrightarrow{z} X(z) \quad \text{ROC} = R_1$$

$$h[n] \xleftrightarrow{z} H(z) \quad \text{ROC} = R_2$$

$$x[n] * h[n] \xleftrightarrow{z} X(z)H(z) \quad \text{ROC} \supseteq R_1 \cap R_2$$

Proof is “identical” to the Fourier/Laplace transform convolution and follows from eigensystem property

Note that pole-zero cancellation may occur between $H(z)$ and $X(z)$ which extends the ROC

While this is true for any two signals, it is particularly important as $H(z)$ represents the **transfer function** of discrete-time LTI system

Example 1: First Order Difference Equation

Calculate the output of a first order difference equation of a input signal $x[n] = 0.5^n u[n]$

$$0.5^n u[n] \xleftrightarrow{z} X(z) = \frac{z}{z - 0.5}$$

System transfer function (z-transform of the impulse response)

$$y[n] - 0.8y[n-1] = x[n]$$

$$H(z) = \frac{1}{1 - 0.8z^{-1}} = \frac{z}{z - 0.8}$$

The (z-transform of the) output is therefore:

$$Y(z) = \frac{z^2}{(z - 0.5)(z - 0.8)} \quad \text{ROC } |z| > 0.8$$

$$= \frac{1}{0.3} \left(\frac{0.8z}{(z - 0.5)} - \frac{0.5z}{(z - 0.8)} \right)$$

$$y[n] = (0.8 * 0.5^n u[n] - 0.5 * 0.8^n u[n]) / 0.3$$

Example 2: 2nd Order Difference Equation

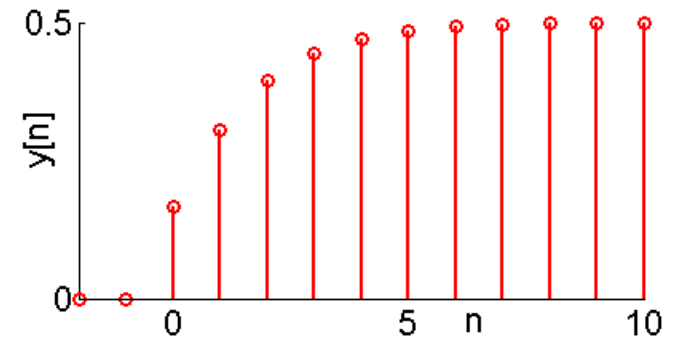
Consider the discrete time step input signal

$$u[n] \xleftrightarrow{z} X(z) = \frac{1}{1-z^{-1}}$$

to the 2nd order difference equation

$$6y[n] - 5y[n-1] + 1y[n-2] = x[n]$$

$$H(z) = \frac{1}{6-5z^{-1}+1z^{-2}} = \frac{1}{(2-z^{-1})(3-z^{-1})}$$



To calculate the solution, multiply and express as partial fractions

$$\begin{aligned} Y(z) &= \frac{1}{(3-z^{-1})(2-z^{-1})(1-z^{-1})} \\ &= 0.5 \frac{1}{(3-z^{-1})} - \frac{1}{(2-z^{-1})} + 0.5 \frac{1}{(1-z^{-1})} \\ &= 0.167 \frac{1}{(1-1/3z^{-1})} - 0.5 \frac{1}{(1-1/2z^{-1})} + 0.5 \frac{1}{(1-z^{-1})} \end{aligned}$$

$$y[n] = (0.167(1/3)^n - 0.5(1/2)^n + 0.5)u[n]$$

Lecture 19: Summary

The z-transform is **linear**

There is a simple relationship for a signal **time-shift**

$$x[n-1] \xleftrightarrow{z} z^{-1} X(z)$$

This is fundamental for deriving the transfer function of a difference equation which is expressed in terms of the input-output signal delays

The **transfer function** of a discrete time LTI system is the z-transform of the system's impulse response

It is a rational polynomial in the complex number z .

Convolution is expressed as multiplication

$$x[n] * h[n] \xleftrightarrow{z} X(z)H(z)$$

and this can be solved for particular signals and systems

Lecture 19: Exercises

Theory

SaS, O&W 10.20

Matlab

Use the inverse z-transform in the symbolic Matlab toolbox to verify Examples 1 and 2 for a first and second order system. See Lecture 18 for the `ztrans()` and `iztrans()` commands

Try changing the coefficients associated with Examples 1 and 2 and verify their behaviour.